

LAB EXPERIMENT 39-40

39)

CODING:

```
import pandas as pd
```

```
import numpy as np
```

```
exam_data = {
```

```
    'name': ['Anastasia', 'Dima', 'Katherine', 'James', 'Emily',
```

```
            'Michael', 'Matthew', 'Laura', 'Kevin', 'Jonas'],
```

```
    'score': [12.5, 9, 16.5, np.nan, 9, 20, 14.5, np.nan, 8, 19],
```

```
    'attempts': [1, 3, 2, 3, 2, 3, 1, 1, 2, 1],
```

```
    'quality': ['yes', 'no', 'yes', 'no', 'no', 'yes', 'yes', 'no', 'no', 'yes']
```

```
}
```

```
labels = ['a', 'b', 'c', 'd', 'e', 'f', 'g', 'h', 'i', 'j']
```

```
df = pd.DataFrame(exam_data, index=labels)
```

```
print("First three rows of the DataFrame:")
```

```
print(df.head(3))
```

OUTPUT:

```
First three rows of the DataFrame:
      name  score  attempts  quality
a  Anastasia   12.5         1     yes
b      Dima    9.0         3      no
c  Katherine   16.5         2     yes
```

40)

CODING:

```
import pandas as pd
import numpy as np

exam_data = {
    'name': ['Anastasia', 'Dima', 'Katherine', 'James', 'Emily',
             'Michael', 'Matthew', 'Laura', 'Kevin', 'Jonas'],
    'score': [12.5, 9, 16.5, np.nan, 9, 20, 14.5, np.nan, 8, 19],
    'attempts': [1, 3, 2, 3, 2, 3, 1, 1, 2, 1],
    'quality': ['yes', 'no', 'yes', 'no', 'no', 'yes', 'yes', 'no', 'no', 'yes']
}

labels = ['a', 'b', 'c', 'd', 'e', 'f', 'g', 'h', 'i', 'j']
```

```
df = pd.DataFrame(exam_data, index=labels)
```

```
print("Selected columns:")
```

```
print(df[['name', 'score']])
```

OUTPUT:

Selected columns:

	name	score
a	Anastasia	12.5
b	Dima	9.0
c	Katherine	16.5
d	James	NaN
e	Emily	9.0
f	Michael	20.0
g	Matthew	14.5
h	Laura	NaN
i	Kevin	8.0
j	Jonas	19.0